Surv 617 HW3

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2025-11-16

The European Social Survey (ESS) is an academically-driven cross-national survey that has been conducted across Europe since its establishment in 2001. For this assignment, we will analyze the ESS-Belgium data collected in 2010. Various formats of the ESS-Belgium data set and the codebook (ESS5_appendix_a6_e03_1.pdf) are all available on Canvas. There are five variables measuring the respondent's trust in leadership:

- TRSTPRL Trust in country's parliament
- TRSTLGL Trust in the legal system
- TRSTPLT Trust in politicians
- TRSTPRT Trust in political parties
- TRSTPLC Trust in the police

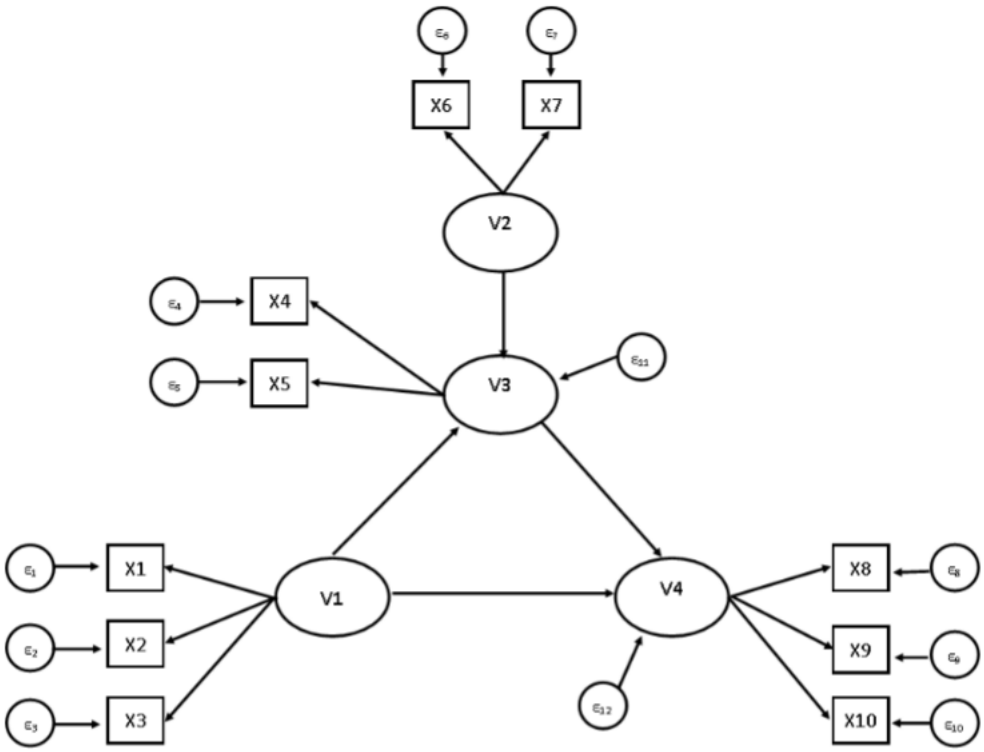
There are three variables measuring attitudes toward immigrants:

- IMSMETN Allow many/few immigrants of same race/ethnic group as majority
- IMDFETN Allow many/few immigrants of different race/ethnic group from majority
- IMPCNTR Allow many/few immigrants from poorer countries outside Europe

And there is one variable measuring the satisfaction with the national government:

- STFGOV How satisfied with the national government

1. Examine the SEM diagram below and answer the following six questions.



a. How many parameters are you estimating?

Each predicted variable (with at least one arrow entering it) has an intercept estimate, a slope estimate for each of its predictors (source of the arrows), and an error variance. The observed variables are only loaded onto one latent variable each. Adding in the error variance and intercept for each, we have $10*(1+1+1) = 30$ parameters (the 1+1+1 signifying the coefficient in front of the latent variables, the intercept, and the error variance). This leaves the endogenous latent variables, V3 and V4, which have 4 associated parameters each (because they are both predicted by two other latent variables, have an intercept, and error variance). This gives a grand total of $30 + 4 + 4 = 38$ parameters in the SEM estimation.

b. Which variables are latent?

V1, V2, V3, and V4 are latent (the errors associated with every variable in the data set are also unobserved (but not latent), there are 12 of them).

c. How many variables are observed?

X1 through X10 are observed, so there are 10 observed variables.

d. Which latent variables are exogenous?

V1 and V2 are exogenous, while V3 and V4 are endogenous.

e. How can you tell whether a latent variable is exogenous using the diagram?

Any variable that is not predicted by any other variable is exogenous, and the arrows in the diagram represent "prediction". So, any latent variables with arrows only *exiting* it are exogenous. Variables have arrows *entering* are endogenous variables, which are all observed variables (X1 through X10) and V3, V4.

f. Suppose that we wanted to add covariance between two latent variables. What would we add to the diagram to indicate this?

A curved line with arrow heads on either side connecting the two latent variables would represent a covariance between them.

2.

Write out the structural and measurement model equations based on the SEM diagram.

Structural Model: (using αs for intercepts)

- $V_3 = \alpha_1 + \gamma_{1,1} V_1 + \gamma_{1,2} V_2 + \epsilon_{11}$
- $V_4 = \alpha_2 + \gamma_{2,1} V_1 + \gamma_{2,2} V_3 + \epsilon_{12}$

Measurement Model: (using βs for intercepts)

- $x_1 = \beta_1 + \lambda_1 V_1 + \epsilon_1$
- $x_2 = \beta_2 + \lambda_2 V_1 + \epsilon_2$
- $x_3 = \beta_3 + \lambda_3 V_1 + \epsilon_3$
- $x_4 = \beta_4 + \lambda_4 V_3 + \epsilon_4$
- $x_5 = \beta_5 + \lambda_5 V_3 + \epsilon_5$
- $x_6 = \beta_6 + \lambda_6 V_2 + \epsilon_6$
- $x_7 = \beta_7 + \lambda_7 V_2 + \epsilon_7$
- $x_8 = \beta_8 + \lambda_8 V_4 + \epsilon_8$
- $x_9 = \beta_9 + \lambda_9 V_4 + \epsilon_9$
- $x_{10} = \beta_{10} + \lambda_{10} V_4 + \epsilon_{10}$

3.

Briefly discuss why it is not advisable to conduct a path analysis on cross-sectional data (i.e., data from a single time point).

Path models are generally more appropriate for data that takes place over time, rather than at a single time point, because the temporal ordering of variables in the causal pathway is important. Without time-ordered variables, researchers would be more likely to observe spurious causal relationships due to potential confounds. The lack of a time element also muddies up inference on causal direction, because no variable strictly preceded or followed any of the others.

4.

Using the ESS dataset, conduct a confirmatory factor analysis (CFA) using the observed variables IMSMETN, IMDFETN, and IMPCNTR as indicators of the latent overall attitude towards immigration. Answer the following three questions:

```r
library(tidyverse)
library(lavaan)
load("C:/Users/jacma/OneDrive/school work/UMD/Surv 617/Data/ess_belgium2.rda")
# recode missing (8) as NA
ess_belgium$imsmetn[ess_belgium$imsmetn==8] <- NA
ess_belgium$imdfetn[ess_belgium$imdfetn==8] <- NA
ess_belgium$impcntr[ess_belgium$impcntr==8] <- NA

fit1 <- cfa(model = 'imm_att =~ imsmetn + imdfetn + impcntr',
            data=ess_belgium) # overall attitude, so no group specification (?)
summary(fit1)$pe
```

```
##       lhs op     rhs exo        est         se        z       pvalue
## 1 imm_att =~ imsmetn   0 1.00000000 0.00000000       NA           NA
## 2 imm_att =~ imdfetn   0 1.28963472 0.03389316 38.05000 0.000000e+00
## 3 imm_att =~ impcntr   0 1.09220316 0.03015712 36.21709 0.000000e+00
## 4 imsmetn ~~ imsmetn   0 0.26927422 0.01149945 23.41627 0.000000e+00
## 5 imdfetn ~~ imdfetn   0 0.08828901 0.01170934  7.54005 4.707346e-14
## 6 impcntr ~~ impcntr   0 0.23550484 0.01147044 20.53146 0.000000e+00
## 7 imm_att ~~ imm_att   0 0.41008121 0.02253198 18.19997 0.000000e+00
```

a.

How is the latent variable being scaled?

The cfa() function fixes one loading for each factor to 1 by default, which we see for the IMSMETN estimate.

b.

Do the estimated loadings seem different from zero, and do they suggest that the observed variables are all a significant function of the latent construct?

The loadings do appear to be different from 0, and have extremely large z-values leading to p-values near 0. This implies they are a significant function of the latent construct. In other words, we see that IMDFETN (tolerance of immigrants of different race/ethic group from majority) and IMPCNTR (tolerance of immigrants from poorer countries outside Europe) are significant indicators of the "overall attitude toward immigration" latent construct, which matches intuition.

c.

Conduct a multiple group analysis to test whether there is measurement invariance by gender (gndr: 1 = male, 2 = female).

```r
fit2 <- cfa(model = 'imm_att =~ imsmetn + imdfetn + impcntr',
            data=ess_belgium,
            group='gndr') # specify gender as group
gender_results <- summary(fit2) # males group 1, females group 2
rbind(gender_results$pe[1:3,c(3,5,7:9)],
      gender_results$pe[11:13,c(3,5,7:9)])
```

```
##        rhs group      est         se        z
## 1  imsmetn     1 1.000000 0.00000000       NA
## 2  imdfetn     1 1.252769 0.04773207 26.24585
## 3  impcntr     1 1.067854 0.04261739 25.05676
## 12 imsmetn     2 1.000000 0.00000000       NA
## 13 imdfetn     2 1.319971 0.04810965 27.43673
## 14 impcntr     2 1.115455 0.04279373 26.06585
```

```r
fit2_const_l <- cfa(model = 'imm_att =~ imsmetn + imdfetn + impcntr',
                    data=ess_belgium,
                    group='gndr', # specify gender as group
                    group.equal = c("loadings")) # ALL loadings equal across fits
anova(fit2,fit2_const_l) # weak invariance
```

```
##
## Chi-Squared Difference Test
##
##              Df    AIC    BIC  Chisq Chisq diff RMSEA Df diff Pr(>Chisq)
## fit2          0 9867.2 9965.0 0.0000
## fit2_const_l  2 9864.3 9951.1 1.0442     1.0442     0       2     0.5933
```

```r
fit2_const_li <- cfa(model = 'imm_att =~ imsmetn + imdfetn + impcntr',
                     data=ess_belgium,
                     group='gndr', # specify gender as group
                     group.equal = c("loadings", # ALL loadings equal across fits
                                     "intercepts")) # constrain intercepts
anova(fit2,fit2_const_li)
```

```
##
## Chi-Squared Difference Test
##
##               Df    AIC    BIC  Chisq Chisq diff RMSEA Df diff Pr(>Chisq)
## fit2           0 9867.2 9965.0 0.0000
## fit2_const_li  4 9863.1 9939.1 3.8451     3.8451     0       4     0.4274
```

```r
fit2_const_liv <- cfa(model = 'imm_att =~ imsmetn + imdfetn + impcntr',
                      data=ess_belgium,
                      group='gndr', # specify gender as group
                      group.equal = c("loadings", # ALL loadings equal across fits
                                      "intercepts", # constrain intercepts
                                      "residuals")) # constrain variances
anova(fit2,fit2_const_liv)
```

```
##
```

```
## Chi-Squared Difference Test
##
##                 Df    AIC    BIC  Chisq Chisq diff RMSEA Df diff Pr(>Chisq)
## fit2             0 9867.2 9965.0 0.0000
## fit2_const_liv   7 9858.2 9917.9 4.9639     4.9639     0       7     0.6644
```

```
fit2_const_livm <- cfa(model = 'imm_att =~ imsmetn + imdfetn + impcntr',
                       data=ess_belgium,
                       group='gndr', # specify gender as group
                       group.equal = c("loadings", # ALL loadings equal across fits
                                       "intercepts", # constrain intercepts
                                       "residuals", # constrain variances
                                       "means")) # constrain means
anova(fit2,fit2_const_livm)
```

```
##
## Chi-Squared Difference Test
##
##                  Df    AIC    BIC  Chisq Chisq diff    RMSEA Df diff Pr(>Chisq)
## fit2              0 9867.2 9965.0  0.000
## fit2_const_livm   8 9865.3 9919.6 14.087     14.087 0.030042       8    0.07953
##
## fit2
## fit2_const_livm .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
rm(fit1,fit2,gender_results,
   fit2_const_l,fit2_const_li,fit2_const_liv,fit2_const_livm)
```

The estimates of loadings for each predictor of the latent attitude seem to be very similar regardless of gender. To test the results more formally, anova() was utilized to complete a Chi-square difference test between the original gender-specific loading model and constrained models. The first constrained model had equal loadings across genders - the resulting test failed to reject the null hypothesis, meaning there is no significant difference in the loadings of the variables toward overall immigration attitudes across gender (the simpler, constrained model is not significantly improved by relaxing the constraint). This allows for a conclusion of "weak" invariance. Constraining the intercepts as well as the loadings also fails to reject the null hypothesis, as does adding in a constraint on the residuals. This may allow us to conclude strong invariance between the groups, however there is a significant difference when constraining the means as well as the other three facets (at $\alpha = 0.1$, at least).

Generally, these findings suggest the relationships (loadings) between the observed variables and our latent construct are the same for males and females. Each of these relationships were statistically significant as well, as seen by the initial model fit.

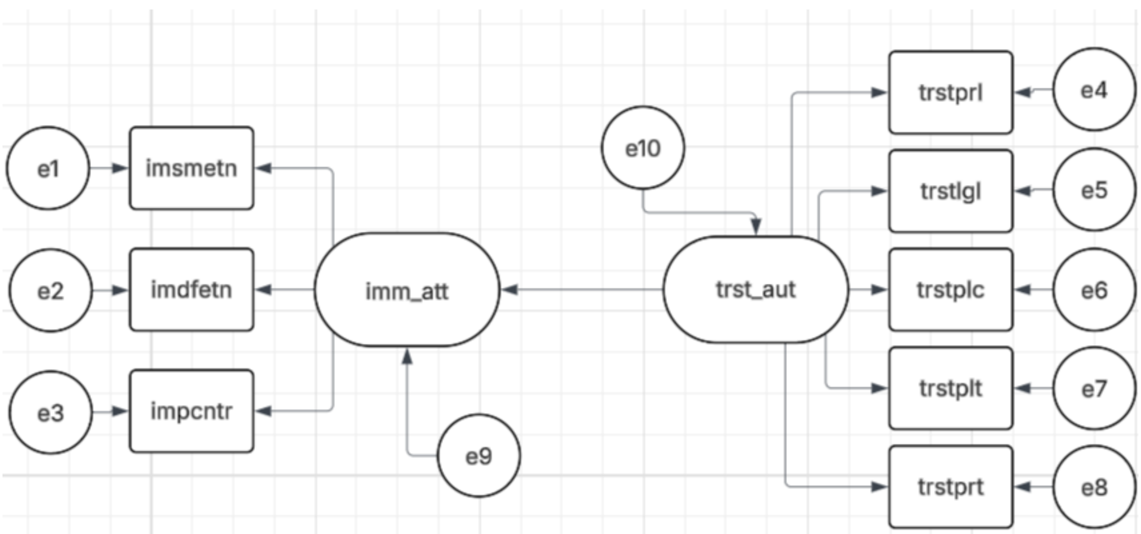
5.

Using the ESS dataset, fit a structural equation model to predict latent immigration attitude (as indicated by the 3 variables in Q4) with a latent measure of trust in authority. The latent measure of trust in authority is indicated by the 5 following observed variables: TRSTPRL, TRSTLGL, TRSTPLC, TRSTPLT, and TRSTPRT.

```r
# recode missing (88) as NA
ess_belgium$trstprl[ess_belgium$trstprl==88] <- NA
ess_belgium$trstlgl[ess_belgium$trstlgl==88] <- NA
ess_belgium$trstplc[ess_belgium$trstplc==88] <- NA
ess_belgium$trstplt[ess_belgium$trstplt==88] <- NA
ess_belgium$trstprt[ess_belgium$trstprt==88] <- NA
model <- '
imm_att =~ imsmetn + imdfetn + impcntr
trst_aut =~ trstprl + trstlgl + trstplc + trstplt + trstprt
imm_att ~ a*trst_aut
'
# an indirect effect would be an "V1 affects V3 through V2" situation (a*b),
# but V1 can also have a DIRECT effect (c) on V3
# total effect = c + (a*b)
fit3 <- sem(model, data=ess_belgium)
results <- summary(fit3,fit.measures=T)
```

6.

Draw the SEM diagram for Q5.



7.

How would you characterize the fit of the SEM from Q5? Acceptable? Poor? Examine modification indices for this model. How could the model be modified to improve fit?

```r
# CFI, RMSEA, AIC/BIC
# modification indices - provide sense of how much the chi-square stat can be
# reduced by estimating additional parameters in the model
results$fit[c(9,17:19,21)]
```

```
##            cfi          rmsea rmsea.ci.lower rmsea.ci.upper   rmsea.pvalue
##      0.9366292      0.1237826      0.1145436      0.1332680      0.0000000
```

```r
modindices(fit3,sort.=T) %>% head(10)
```

```
##        lhs op     rhs      mi    epc sepc.lv sepc.all sepc.nox
## 55 trstplt ~~ trstprt 400.952  1.771   1.771    2.153    2.153
## 50 trstlgl ~~ trstplc 255.227  1.295   1.295    0.413    0.413
## 46 trstprl ~~ trstlgl 103.238  0.758   0.758    0.275    0.275
## 51 trstlgl ~~ trstplt  68.222 -0.513  -0.513   -0.323   -0.323
## 20 imm_att =~ trstprl  48.559 -0.468  -0.300   -0.134   -0.134
## 52 trstlgl ~~ trstprt  42.487 -0.402  -0.402   -0.238   -0.238
## 54 trstplc ~~ trstprt  37.352 -0.336  -0.336   -0.207   -0.207
## 21 imm_att =~ trstlgl  33.240 -0.449  -0.287   -0.124   -0.124
## 49 trstprl ~~ trstprt  32.773 -0.358  -0.358   -0.252   -0.252
## 53 trstplc ~~ trstplt  24.727 -0.274  -0.274   -0.180   -0.180
```

```r
rm(model,fit3,results,ess_belgium)
```

AIC and BIC could be used as comparative metrics if another model had been fit, but interpretation of them on their own is meaningless. The RMSEA was estimated to be 0.123, with a narrow 90% confidence interval and thus a very small p-value for the test of it being less than or equal to 0.05. This leads us to conclude that the model did not fit the data very well, because the implied covariances of the model are not particularly close to the sample covariances. The CFI is 0.936, which is a desirable value and indicates the fitted model at least does much better than a null model. The RMSEA and CFI almost seem contradictory, but the former seems to indicate that more work is needed (because a large CFI seems to be a natural by-product - *some* structure is expected to be better than pure random chance).

The modification indices show that there are ways to improve the model fit. There are several extremely large indices (10 or greater, some well into the hundreds), and all of them are either covariance terms (e.g. a parameter to capture the relationship *between* trstplt and trstprt - expected to have the largest standardized effect on the parameters (sepc.all)), or to add additional loadings to the latent factors (e.g. trstprl loaded onto imm_att). All of the top indices relating to residual covariance were between the trust variables). In a diagram, this would be represented by curved arrows connecting every item in the space. It seems natural that this would improve model fit, as it allows for the capturing of additional sources of variability within each variable, and thus improves the quality of the loadings on the latent factors. However, there is a point where such a model becomes uninterpretable or unintuitive, which further models would have to keep in mind. The theory driving the structural equation model is the key aspect, and one ought to be careful to avoid "overfitting" the data and disregarding the theory they started with just to improve performance. Overall, the decision on whether the fit is "acceptable" or "poor" is dependent upon what the researcher wants to achieve. If they are looking for the "most" accurate solution, this model certainly is not satisfactory. However, it may be acceptable if the model seems to be the most appropriate set-up in *theory*.

8.

In ~1/2 a page, describe what the model from Q5 tells you about the relationship between trust in authority and immigration attitudes. Are there conclusions you are not able to make (using only this model and these data)?

The model gives a regression estimate of -0.102 between the latent factor for trust in authority and the latent factor for attitudes toward immigration. The estimate was statistically significant as well, with a large (in absolute value) z-value. This indicates that increased trust in authority leads to a "diminished" or more negative attitude toward immigrants. There are a plethora of other conclusions that cannot be made with this model, such as the effects of any confounds such as location or residence of the survey respondents. One may expect there to be a difference between urban and rural groups in their attitudes toward immigration and their trust in authority, along with the relationship between these two. The current model specification also does not give any analysis on correlation between any of the observed variables, which seem to be a natural inclusion because some relationships seem obvious (e.g. one might expect trust in political parties and trust in politicians to be highly correlated). The findings are also inherently limited until they are applied to additional data sets, perhaps from other countries and/or other times. If similar results to those found here are seen in other locations or at other times, the conclusions about the relationship between the factors *and* the strength of the indicators on each would be augmented significantly. \ On a more general note, the inherent difficulty with SEM (in this context) is that we are trying to develop a way of understanding some latent, *unobserved* concept. The quality (accuracy, reliability, representativeness) of the variables loaded onto these factors is naturally critical as a result.